Permutations

- In how many ways can we select three students from a group of five students to stand in line for a photo?
- Note: *the order in which the students are selected matters* because it changes the order in the photo

- **Solution:**
  There are 5 ways to select the first student to stand at the start of the line. Afterwards, there are 4 ways to select the student to stand second in the line from the remaining 4 students. Finally, there are 3 ways to select the third student in line. By the product rule, there are a total of $5 \times 4 \times 3 = 60$ ways